Quantized ResNet-18 Acceleration on RV64IMC with Custom ISA Extensions



114062512 柯哲元

Bare-Metal Baseline

Full-model W8A8 ResNet-18 on RV64IMC Spike

Baseline Configuration

Model	**W8A8 ResNet-18**
ISA	**RV64IMC**
Execution	**Bare-metal**
Simulator	**Spike**
Graph	**37 operators**

Integer-only kernels • static memory • no floating-point hardware

Full-Model Correctness

✓ PASS

1000 outputs

mismatches = 0

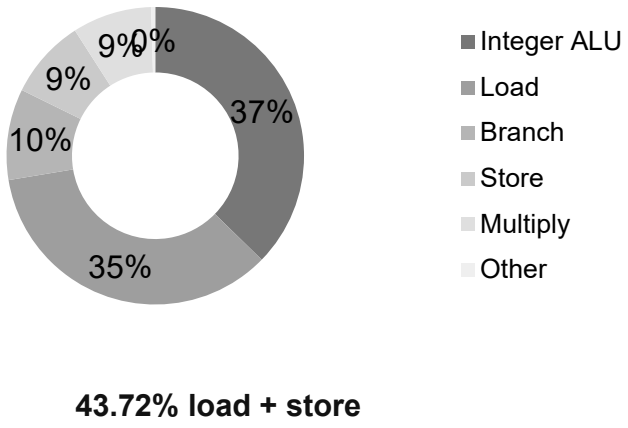
All exported operator outputs match the software reference.

Baseline Profiling & Bottlenecks

Spike profiling shows a convolution-dominated, memory-heavy scalar workload

20.239B	8.848B	0.0896	98.64%
Retired Instructions	Memory Instructions	Dense MAC / Inst.	CONV_2D Share

Instruction Mix



43.72% load + store

Operator Contribution



Memory Overhead

43.72% of retired instructions are loads/stores.

Low MAC Efficiency

0.0896 MAC/inst ≈ 11.16 instructions per MAC.

Loop Overhead

Branches account for 9.90% of retired instructions.

XQMAC8 — 8-Way Packed INT8 MAC

Targeting the dominant convolution inner loop

Instruction Semantics

```
rs1   [A7 A6 A5 A4 A3 A2 A1 A0]

rs2   [W7 W6 W5 W4 W3 W2 W1 W0]
```

✖

$$rd = rd + \Sigma\, Ai \times Wi$$

Packed 64-bit operands
8 signed INT8 MACs / instruction

Measured Impact (B0 → X1)

Dynamic Instructions	20.239B	4.648B	−77.03%
Dense MAC / Instruction	0.0896	0.3903	4.35×
Memory Instructions	8.848B	1.266B	−85.69%

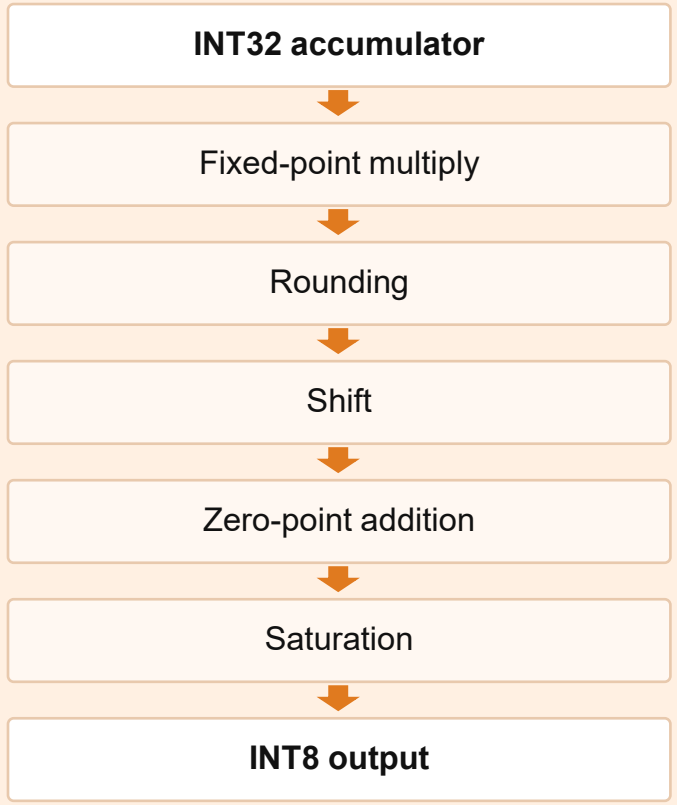
XQMAC8 MAC Coverage 92.99%

196.64M XQMAC8 instructions → 1.573B packed MACs

XREQUANT — Fused Requantization

Incremental evaluation on top of XQMAC8

One Instruction Replaces



Incremental Effectiveness (X1 → X2)

Metric	X1	X2	Change
ADD inst/output	103.04	65.87	−36.08%
Store	10.45M	5.49M	−47.47%
Integer ALU	2.010B	1.915B	−4.71%
Branch	704.67M	688.59M	−2.28%

End-to-end: 4.648B → 4.696B (+1.04%)

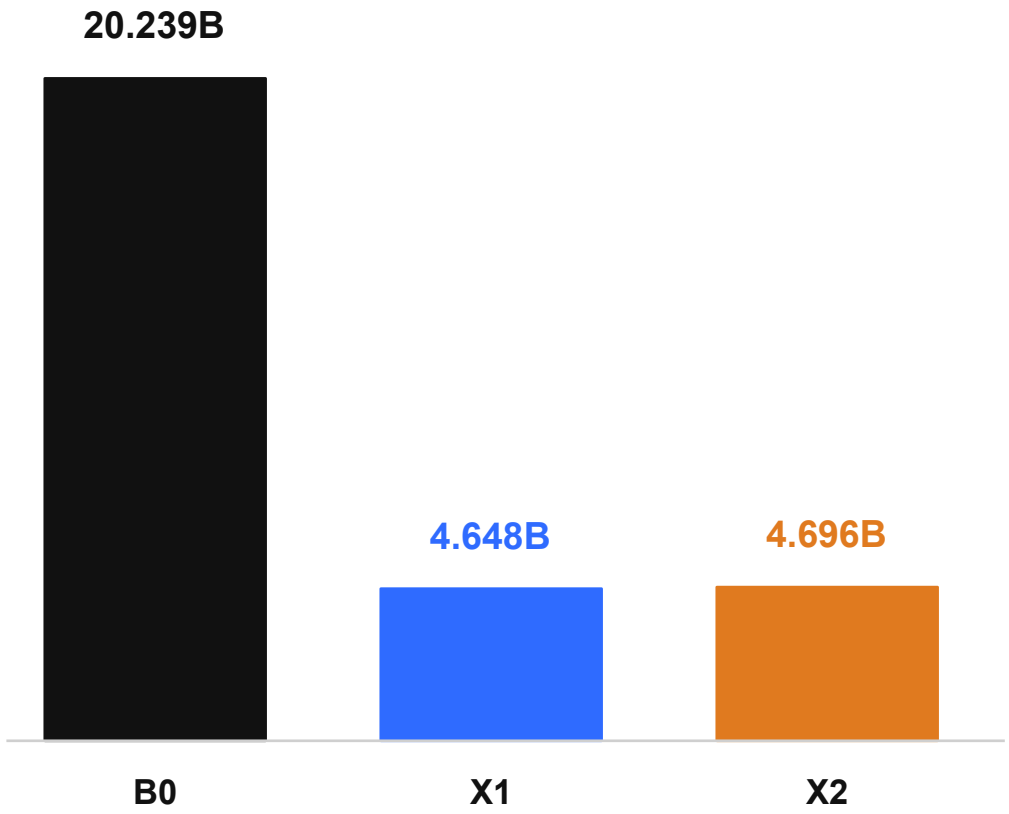
Extra loads in the first RGB convolution offset the local savings.

3.236M XREQUANT executions — one per accelerated Conv/ADD output

Final Results & Takeaways

ISA-level evaluation on Spike

Full-Model Retired Instructions



Overall X2 vs B0: 76.80% fewer retired instructions

Key Takeaways

1 **Baseline**

Full-model W8A8 ResNet-18 runs correctly on RV64IMC Spike.

2 **XQMAC8**

Main performance gain: −77.03% retired instructions.

3 **XREQUANT**

Local requantization benefit verified; Conv integration still needs refinement.

Next: cycle-aware RTL evaluation and other optimization.